

credit_card_fraud.ipynb

Interactive Jupyter Notebook — End-to-End Fraud Detection

■ Overview

This Jupyter Notebook provides a complete, interactive, end-to-end walkthrough of a Credit Card Fraud Detection pipeline. It combines data downloading, preprocessing, model training, and evaluation in a single runnable document. It is the 'all-in-one' version of the separate *download_data.py* and *model.py* scripts.

■ Libraries Used

- **urllib.request** — downloads the dataset directly from a public URL
- **pandas, numpy** — data manipulation
- **scikit-learn** — ML model, metrics, train/test split
- **imbalanced-learn (SMOTE)** — handles class imbalance
- **matplotlib, seaborn** — visualisations rendered inline in the notebook

■ Notebook Cells — Explained

Cell 1 — Title

- A markdown heading: "Credit Card Fraud Prediction using Random Forest & SMOTE". Sets the context for the notebook.

Cell 2 — Imports & Data Download

- Imports all required libraries.
- Defines `DATA_FILE = 'creditcard.csv'` and `DATA_URL` pointing to a public TensorFlow storage bucket.
- If the CSV does not exist locally, `urllib.request.urlretrieve()` downloads it automatically.
- Loads the CSV into a pandas DataFrame and displays `df.head()` — the first 5 rows.

Cell 3 — Preprocess & Train & Evaluate

- Drops NaN rows.
- Separates features (X) and labels (y = 'Class').
- Splits 80/20 into train/test with stratification.
- Applies SMOTE to the training set to balance fraud vs. legitimate samples.
- Trains `RandomForestClassifier(n_estimators=50, random_state=42, n_jobs=-1)`.

- Predicts on the test set.
- Prints a classification report.
- Renders a Confusion Matrix heatmap inline using seaborn.

■ Difference from model.py

Data download	Notebook downloads data automatically via URL; model.py expects the file to already exist
Visualisation	Notebook shows plots inline (plt.show()); model.py saves them as PNG files
Entry point	Notebook cells run top-to-bottom interactively; model.py runs via command line
PR Curve	model.py also plots a Precision-Recall curve; the notebook currently only shows the Confusion Matrix

■■ How to Run

Open the notebook in JupyterLab or Jupyter Notebook:

```
jupyter notebook lithish/credit_card_fraud.ipynb
```

Then click **Run All Cells**. The data will download automatically on first run (~150 MB). Training takes 1–3 minutes depending on your CPU.

■ Tips

- Run Cell 2 first; wait for the download to finish before running Cell 3.
- The SMOTE step can take 30–60 seconds on the full ~285k-row dataset.
- Change `n_estimators` to 100 or 200 for potentially higher accuracy at the cost of longer training.